



Stalled in space

By the start of the 21st century, human space flight had not progressed as rapidly as might have been expected, given the achievements of the first decade of space travel. Thirty years after the last man stood on the surface of the Moon, there was no sign of a resumption of such long-distance manned space journeys, despite the efforts of enthusiasts such as former astronaut Buzz Aldrin to drum up support for missions to Mars and beyond. What is more, the comparatively simple goal of a safe, fully reusable “space liner” to fly to space stations and back had remained elusive. The second Shuttle disaster, when *Columbia* broke up on re-entry on February 1, 2003, was a massive setback for NASA. It once more saw the space agency accused of sloppiness in its safety procedures and placed a question mark against continuation of Shuttle flights. This in turn put the whole future of the ambitious International Space Station (ISS) project at risk.

Fresh vision

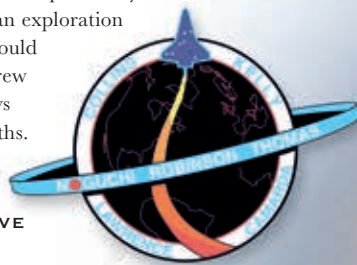
At the root of the problems threatening to turn human space flight into a historical dead end was the failure to identify a clear purpose that would guarantee consistent political and financial support. In the 1960s manned missions had a clear goal for both the Americans and the Soviet Union – to be the winner in a race. But once the competition came to an end in the 1970s, the decisive incentive was lost.

The sense of drift in the American manned space program was brought to an end on January 14, 2004, when President George W. Bush announced that he was giving NASA “a new focus and vision for future exploration.” The president stated: “We will build new ships to carry man forward into the Universe, to gain a new foothold on the Moon and to prepare for new journeys to the worlds beyond our own.” Initial reactions suggested that around half the American people were sceptical about the new vision, but NASA reacted with enthusiasm.

COLUMBIA BREAK-UP

Disintegration of the Space Shuttle Columbia high in the atmosphere on February 1, 2003, resulted in the loss of seven crew on board. The disaster was not only a human tragedy, but also a severe setback for NASA's ongoing manned space program. The Shuttle fleet was grounded for the next two and a half years.

The completion of the ISS remained a major concern. The delays of the Shuttle, relaunched in 2005, but finally retired in 2011, slowed the building of the ISS although the project continued regardless. Two replacements for the Shuttle were developed, the SpaceX Crew Dragon and the Boeing CST-100 Starliner, both intended to ferry crew to the ISS. In the wake of President Bush’s “vision” speech, however, the overriding goal for NASA became the return to the Moon and, beyond that, a manned mission to Mars. Conscious of cost limitations, they banked on as much reuse of existing technology as was feasible. NASA’s new reusable spacecraft, the Orion Multi-Purpose Crew Vehicle, was an improved, updated version of the Apollo capsule. Specifically designed for human exploration of deep space, it could accommodate a crew of four for journeys lasting many months.



COMMEMORATIVE BADGE

This badge bears the names of the seven astronauts who took part in the Shuttle return-to-flight mission in July 2005: Eileen Collins, Jim Kelly, Soichi Noguchi, Steve Robinson, Wendy Lawrence, Charlie Camarda, and Andy Thomas.

SHUTTLE RELAUNCH

Space Shuttle Discovery launches from NASA's Kennedy Space Center on July 21, 2005 on the first mission after the loss of Columbia. Unfortunately, the mission revealed that the problem with debris from the external fuel tank striking the orbiter during the launch – the cause of the Columbia disaster – had not been solved.

To launch Orion, NASA planned to use a Shuttle-derived Space Launch System (SLS), which would be the most powerful rocket ever made. Delays inevitably occurred. NASA originally intended a human space flight to the Moon by 2018, but this later became the projected date for the first uncrewed Orion mission, with the Moon landing put back to the 2020s. At some time after that date a permanent moonbase was to be set up, possibly exploiting resources such as frozen water that the apparently barren satellite might have to offer.

And so to Mars

The Orion spacecraft was specifically designed with a potential Mars voyage in view, but the challenge of establishing a human base on the red planet was

